

Automated Traffic Incident Summarisation Using LLMs: A Smart City Perspective

Rajeev Ranjan Pandey*, Khalid Elgazzar†*

*20250007918@students.cud.ac.ae, †khalid.elgazzar@ontariotechu.ca

*Canadian University Dubai, UAE

†IoT Research Laboratory, Ontario Tech University, Canada

Abstract—Traffic control centers process incident reports that are often too long for rapid operational use. This paper evaluates transformer-based summarization for converting such reports into concise, decision-ready briefings. Using a dataset of over 5,250 records derived from the US Accidents dataset and structured Dubai Pulse traffic data converted into narrative descriptions, BART-large-cnn, Flan-T5-small, and PEGASUS-CNN/DailyMail are compared with TextRank under zero-shot settings. BART achieves the strongest overall performance with ROUGE-1 of 0.432, ROUGE-2 of 0.198, and ROUGE-L of 0.391, outperforming TextRank by 35.8% on ROUGE-1. However, qualitative analysis shows that strong lexical scores do not ensure operational reliability: BART hallucinates safety-critical details in approximately 14% of outputs, while Flan-T5 omits road-closure information in approximately 11%. These findings show that traffic incident summarization systems for smart city operations should be evaluated using both lexical overlap and factual consistency.

Keywords—traffic incident summarization, abstractive summarization, transformer models, zero-shot evaluation, ROUGE metrics, factual hallucination, smart city operations, GCC traffic data, safety-critical NLP, extractive-abstractive comparison

I. INTRODUCTION

Smart cities generate continuous streams of traffic incident reports for control rooms, dispatch, and mobility monitoring. These reports contain useful operational details but are too long for fast decision-making. Dispatchers must scan lengthy narratives to find the incident type, affected segment, closure status, and severity. In dense urban systems, this delay creates friction between information availability and response action.

Automatic summarization can reduce this gap between report length and decision-critical content. Extractive methods are computationally simple and preserve source wording, but they often fail when important details are distributed across multiple sentences. Abstractive transformer models generate more coherent summaries by synthesizing information across the document, but they may also hallucinate unsupported facts, which is risky in safety-critical traffic operations.

This paper makes three contributions. (1) A traffic incident summarization dataset of over 5,250 records is constructed by combining US Accidents [4] and Dubai Pulse [5] data, providing GCC regional grounding. (2) A zero-shot evaluation of BART, Flan-T5, and PEGASUS against TextRank is presented using both ROUGE and qualitative failure analysis. (3) A comparison tool is developed to evaluate summarization outputs in an applied smart mobility context.

II. RELATED WORK

Transformer-based summarization models have significantly improved abstractive text generation over earlier sentence-ranking approaches. BART [1] uses a denoising sequence-to-sequence pretraining objective in which corrupted text is reconstructed, enabling strong contextual modeling across long inputs. T5 [2] formulates all NLP tasks in a unified text-to-text paradigm, which makes instruction-style prompting natural for summarization. PEGASUS [3] introduces gap-sentence generation, a pretraining objective specifically designed for summarization tasks, and has shown strong performance on news benchmarks.

TextRank remains an important extractive baseline. It ranks sentences using graph centrality derived from pairwise sentence similarity. It offers a useful reference because it preserves factual wording from the source and avoids generation-specific errors such as hallucination.

A key limitation of abstractive summarization is factual reliability. Maynez et al. showed that strong ROUGE performance does not guarantee faithfulness, as models may produce summaries with high n-gram overlap while still introducing unsupported details. This issue is especially important in traffic operations, where incorrect closure, injury, or incident status information may affect dispatch decisions, rerouting logic, and public communication.

Proposed Approach. Based on these gaps, this study adopts a zero-shot framework evaluating BART, Flan-T5, and PEGASUS against TextRank on traffic reports. In addition to ROUGE metrics, qualitative failure analysis captures hallucination, omission, and over-compression errors. This makes the evaluation more suitable for safety-critical operations, where lexical overlap alone is insufficient to judge deployment readiness.

III. METHODOLOGY

A. Dataset Construction

The corpus combines over 5,250 records from the US Accidents dataset [4] with GCC data from Dubai Pulse, UAE Federal Traffic Statistics, and Abu Dhabi Open Data [5]. Because GCC portals mainly provide structured records, these were normalized and converted into operator-style narrative incident reports using a rule-based text generation approach. This resulted in 250 GCC narrative samples aligned with operational reporting formats.

ROUGE evaluation used a stratified set of 420 records (300 US, 120 GCC) across five incident categories: rear-end collisions, highway lane blockages, weather-related events, pedestrian incidents, and multi-vehicle pile-ups. Each record was paired with a human-written reference summary of 40 to 80 tokens, allowing both quantitative and qualitative comparison across models.

B. Models and Configuration

BART-large-cnn [1], Flan-T5-small [2], and PEGASUS-CNN/DailyMail [3] were evaluated in zero-shot mode. BART was selected because incident reports share some structural similarity with news-style event descriptions. Flan-T5 was included as a lighter instruction-oriented model that supports prompt-based summarization through the `summarize:` prefix. PEGASUS was used as a summarization-specialized model trained with a gap-sentence objective, providing an additional abstractive comparison point. TextRank served as the extractive baseline.

Inputs were truncated to 512 tokens. Shared generation parameters were: beam size 2, length penalty 1.0, max-new-tokens 72, min-new-tokens 18, and no-repeat-ngram-size 3. These settings were chosen to encourage compact summaries while reducing repetitive generation. ROUGE metrics were computed using the rouge-score library with stemming. To evaluate the reliability of each model for real-time deployment, execution times were also recorded on a standard cloud GPU environment.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

Table I presents ROUGE scores, compression ratios (CR), execution times, and ROUGE-1 gains over TextRank. All three abstractive models outperform the baseline. BART achieves the strongest overall results (ROUGE-1: 0.432, ROUGE-2: 0.198, ROUGE-L: 0.391), representing a 35.8% improvement over TextRank.

TABLE I
ROUGE SCORES (R-1/R-2/R-L), COMPRESSION RATIO (CR), ROUGE-1 GAIN VS. TEXTRANK ($n = 420$), AND EXECUTION TIME

Model	R-1	R-2	R-L	CR	Gain R-1	Exec.(s)
Abstractive Models						
BART-large-cnn	0.432	0.198	0.391	3.2x	+35.8%	1.42
Flan-T5-small	0.408	0.181	0.372	3.5x	+28.3%	0.28
PEGASUS-CNN/DM	0.389	0.162	0.354	3.8x	+22.3%	1.65
Extractive Baseline						
TextRank	0.318	0.109	0.287	2.8x	—	0.05

$R-1 = \text{ROUGE-1}$, $R-2 = \text{ROUGE-2}$, $R-L = \text{ROUGE-L}$, $CR = \text{Compression Ratio}$, $\text{Exec.}(s) = \text{Exec. Time per record}$.

All abstractive models outperform TextRank, indicating that synthesis across sentences improves summary quality for traffic reports. BART performs best because incident reports share structural characteristics with event-oriented news text. However, for real-time traffic applications, execution time is a critical factor. While BART offers the highest summarization

quality, it requires approximately 1.42 seconds per record. In contrast, Flan-T5 achieves competitive results while operating much faster (0.28 seconds), offering a highly reliable trade-off for time-sensitive deployments. PEGASUS produces shorter headline-like outputs that may be less aligned with multi-fact traffic reporting requirements.

Error Analysis. A qualitative review of a stratified sample showed that BART introduces hallucinated details in approximately 14% of outputs. For example, it occasionally fabricates injury counts (e.g., stating "3 injuries" when the source only reports a minor collision). To mitigate this, operators could use constraint-based decoding or post-generation factuality checks. Flan-T5 omits road closure information in approximately 11% of cases. PEGASUS tends to over-compress summaries, removing location or response details. TextRank avoids hallucination but fails to capture information when the most important facts are distributed across multiple sentences.

These results demonstrate that ROUGE alone is insufficient for evaluating summarization in safety-critical systems. A model that scores well lexically may still omit or invent critical operational details. Therefore, future assessments should incorporate semantic metrics such as BERTScore or similar methods, which evaluate underlying meaning rather than strict n-gram overlap. This also motivates exploring hybrid extractive-abstractive designs that balance factual grounding and fluency.

V. CONCLUSION

This study evaluates transformer-based summarization models for traffic incident reports in a smart mobility setting. While BART achieves the best ROUGE performance, qualitative analysis highlights limitations in factual reliability that remain important for deployment-oriented use. These findings suggest that future systems should combine extractive grounding with abstractive generation, alongside stronger factuality checks. Future work includes domain-specific fine-tuning, Arabic-language support for GCC deployments, and larger-scale evaluation across more geographically diverse incident corpora.

CODE AVAILABILITY

The source code and dataset are available on GitHub (VizArtPandey/CUD_AAI_MINIPROJECT_LLMs-for-Traffic-Incident-Summarization). A live demo is accessible at Hugging Face Spaces (rajvivan/CUD-Traffic-AI).

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